

ADAM CATTO

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EXPERIENCE

Data Scientist, Icahn School of Medicine at Mount Sinai

Jun 2022 – Present

- First author on a deep-learning-driven behavioral neuroscience manuscript; built an end-to-end pipeline combining active-learning pose estimation with time-series models (LSTM, TFT) to forecast near-term behavioral events.
- Designed and trained from scratch novel deep learning models for DNA sequence foundation models. Re-implemented and train from scratch Google DeepMind's AlphaGenome foundation model in PyTorch/DeepSpeed.
- Designed a novel deep learning task for neurogenomics, predicting cell-type-specific gene expression from personal genome sequence variants. Implemented and validated on Alzheimer's Disease (AD) dataset and discovered genetic drug targets for AD.
- Designed and trained a multi-scale 1D CNN for detecting neuron spikes in wavelet-decomposed electrophysiology signals, achieving an F1 of 81% and eliminating dozens of hours of manual annotation per experiment.
- Developed a multimodal few-shot segmentation and pose-tracking pipeline using Meta AI's foundation models and Grounded-SAM-2, including a Streamlit interface for remote labeling. Built segmentation-guided keypoint inference and scaled the system to process 200+ hours of multichannel animal-behavior video on an NVIDIA H100 cluster.
- Implemented a local LLM-based question-answering RAG agent with Llama 3.2 70B over thousands of neuroscience papers.
- Led analysis of spatial transcriptomics, whole-brain microscopy, animal behavior, and electrophysiology datasets. Co-author on multiple manuscripts.
- Benchmarked self-supervised deep learning algorithms for medical imaging tasks.

Research Data Scientist Intern, Johnson & Johnson

Jun 2021 – Oct 2021

- Discovered 9 digital biomarkers of 4 autoimmune disorders from wearable sensor data.
- Implemented distributed pipeline for terabyte-scale accelerometer data. Improved running time > 100x.
- Implemented signal processing algorithms to detect interpretable sleep and activity patterns in accelerometer data.
- Trained machine learning models to classify disease vs. healthy controls from interpretable time-series features.

Graduate Research Assistant, Research Foundation of CUNY

Aug 2020 – Jun 2022

- Contributed to data preprocessing, machine learning model building, and visualization/analysis that won a national data challenge hosted by the National Institutes of Health: <https://www.nichd.nih.gov/research/supported/challenges/decoding-maternal-morbidity>. Wrote multiple technical machine learning preprints documenting the methods and results.
- Conceptualized, implemented, and validated two novel ensemble learning techniques for handling large amounts of missing data in tabular datasets, with and without imputation.
- Built POMDP-based sequential decision making clinical decision support system to optimize pregnancy treatment plans to avoid adverse outcomes in pregnancy

SKILLS

- **Languages:** Python (Advanced), Shell Scripting, R, JavaScript
- **Machine Learning / Data Analysis:** PyTorch, PyTorch-Lightning, PyTorch-XLA, DeepSpeed, TorchAudio, TorchVision, PyTorch-Forecasting, OpenCV, Pandas, Scikit-Learn, HuggingFace, LlamaIndex, LangChain, RAG
- **Miscellaneous:** Linux, Google Cloud Platform, Large-Scale Deep Learning, Streamlit, LSF, Git

EDUCATION

CUNY Graduate Center, MS Data Science

2020 – 2022

- **(GPA 3.90/4.00) Graduate Coursework:** Machine Learning, Data Visualization, Artificial Intelligence, Data Mining, Computational Biology, Digital Image Processing, Deep Learning for Genomics, Big Data Analytics
- **Thesis:** *Hierarchical Model Transfer Methods for Ensemble Learning with Large Amounts of Missing Data*

Stony Brook University, BS Applied Mathematics & Statistics, Philosophy

2015 – 2019

PROJECTS

AI Bioinformatics Research Assistant

Used LlamaIndex, Ollama, and Llama3.2 to create a local RAG agent that autonomously plans and executes an analysis of next-generation biological sequencing data, demonstrating its reasoning and outputs in real-time in a Jupyter notebook.

RoomReport

Wrote a React Native mobile app for apartment building managers to create inspection reports.